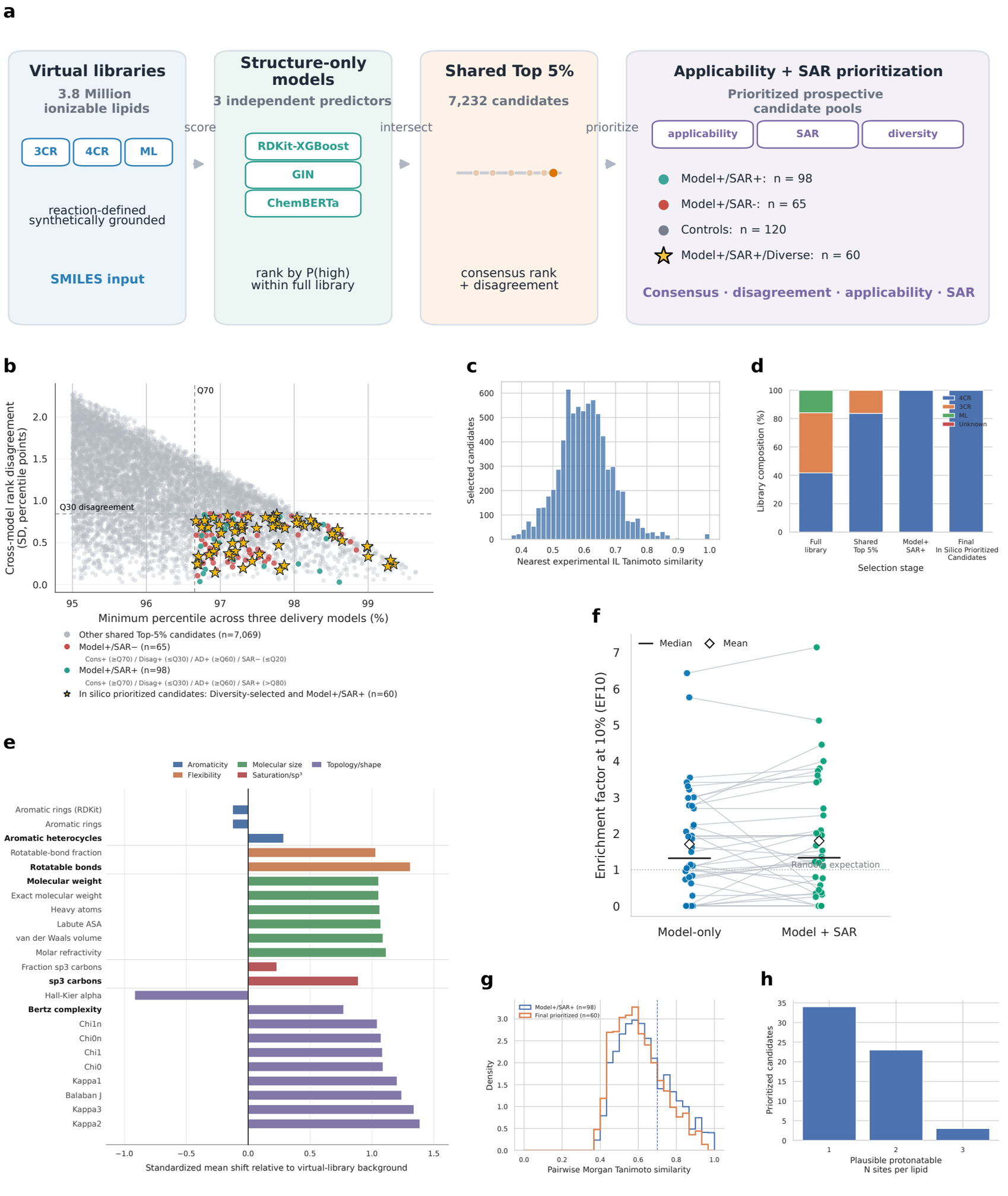

a
Virtual libraries
3.8 Million ionizable lipids
3CR
4CR
ML
reaction-defined synthetically grounded
SMILES input
score
Structure-only models
3 independent predictors
RDKit-XGBoost
GIN
ChemBERTa
rank by P(high) within full library
intersect
Shared Top 5%
7,232 candidates
consensus rank + disagreement
prioritize
Applicability + SAR prioritization
Prioritized prospective candidate pools
applicability
SAR
diversity
Model+/SAR+: n = 98
Model+/SAR-: n = 65
Controls: n = 120
Model+/SAR+/Diverse: n = 60
Consensus · disagreement · applicability · SAR
b
Cross-model rank disagreement (SD, percentile points)
Q70
Q30 disagreement
Minimum percentile across three delivery models (%)
Other shared Top-5% candidates (n=7,069)
Model+/SAR− (n=65)
Cons+ (≥Q70) / Disag+ (≤Q30) / AD+ (≥Q60) / SAR− (≤Q20)
Model+/SAR+ (n=98)
Cons+ (≥Q70) / Disag+ (≤Q30) / AD+ (≥Q60) / SAR+ (>Q80)
In silico prioritized candidates: Diversity-selected and Model+/SAR+ (n=60)
c
Selected candidates
Nearest experimental IL Tanimoto similarity
d
Library composition (%)
4CR
3CR
ML
Unknown
Full library
Shared Top 5%
Model+ SAR+
Final In Silico Prioritized Candidates
Selection stage
e
Aromaticity
Molecular size
Topology/shape
Flexibility
Saturation/sp³
Aromatic rings (RDKit)
Aromatic rings
Aromatic heterocycles
Rotatable-bond fraction
Rotatable bonds
Molecular weight
Exact molecular weight
Heavy atoms
Labute ASA
van der Waals volume
Molar refractivity
Fraction sp3 carbons
sp3 carbons
Hall-Kier alpha
Bertz complexity
Chi1n
Chi0n
Chi1
Chi0
Kappa1
Balaban J
Kappa3
Kappa2
Standardized mean shift relative to virtual-library background
f
Median
Mean
Enrichment factor at 10% (EF10)
Random expectation
Model-only
Model + SAR
g
Model+/SAR+ (n=98)
Final prioritized (n=60)
Density
Pairwise Morgan Tanimoto similarity
h
Prioritized candidates
Plausible protonatable N sites per lipid